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Course Code : CSA 14 06

Course : Compiler Design

C 0 2

Assessment Tool - 3

Memories Based Task

27) (a) Compiler processes large programs with syntax errors;

Input :

id ++ id + id

Grammar

$E \rightarrow E + T \mid T$

$T \rightarrow id$

(i) Identify error positions :

Error occurs at the second '+' because two operators appear consecutively.

id + _^ + id + id

(ii) Apply Parse Mode Recovery

- * Detect the unexpected +.
- * Skip the invalid +.
- * Continue parsing from the next id.

Revised input :

id + id + id

(iii) Demonstrate parsing continuation

E
 $\rightarrow E + T$
 $\rightarrow E + T + T$
 $\rightarrow T + T + T$
 $\rightarrow id + id + id$

(iv) Explain synchronization tokens :

synchronization tokens : id, +, \$

The parser skips invalid tokens until one of these tokens is found and then continues parsing.

(v) Suggested improvements :

* Better error messages.

* Automatic correction suggestions

* Report multiple errors.

* Use phrase-level or phrase-ruled recovery.

(b) IDE supports embedded scripting :

Example :

id = id + id

Grammar :

$S \rightarrow id = E$

$E \rightarrow E + T \mid T$

$T \rightarrow id$

(i) Explain integration of parsing levels :

⇒ lexical analysis : Identifies tokens (id, =, +).

⇒ syntax analysis : Checks grammar.

⇒ semantic analysis : Checks variable declarations and types.

(ii) Demonstrate parsing :

S

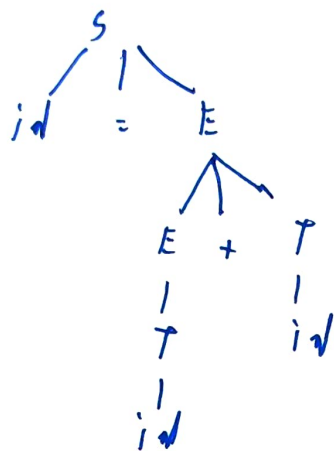
$\rightarrow id = E$

$\rightarrow id = E + T$

$\rightarrow id = T + T$

$\rightarrow id = id + id$

(iii) Construct parse tree :



(iv) Identify error :

For the given input $id = id + id$, no syntax error is present.

Possible errors in embedded scripting :

- Missing =
- Missing operand
- Invalid identifier

(v) Suggest improvements :

(i) Better syntax highlighting

(ii) Auto - completion.

(iii) Real - time syntax checking.

(iv) Error recovery.

(v) Mixed - language parser support.

(c) AI - based code revision tool

Grammar :

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid id$$

Input :

$$id + (id * id$$

(i) Explains how the parser identifies the error

The parser expects a closing parenthesis), but the input ends.

Error :

$$id + (id * id$$

missing ')'

(ii) Demonstrates parsing until failure

$$\begin{aligned} E & \rightarrow E + T \\ & \rightarrow T + T \end{aligned}$$

$$\rightarrow id + T$$

$$\rightarrow id + T * F$$

$$\rightarrow id + F * F$$

$$\rightarrow id + (E * id)$$

Parsing fails because $)$ is missing.

(iii) suggest possible corrections:

Correct input:

$$id + (id * id)$$

or insert the missing $)$ before the end of input

(iv) Explain how the system predicts next valid token

After parsing $(id * id$, the parser expects:

$)$

It predicts $)$ using the grammar and parser state.

(V) Propose an error recovery mechanism:

⇒ Use panic - mode recovery.

⇒ Insert the missing) automatically.

⇒ Continue parsing after recovery.

⇒ Display an error message with the location and suggested fix.